

Morning Paper Report - Friday, July 03, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Chess expertise improves rule-guided flexibility and visual working memory precision

Marjan Makhsous, Mohammad Jowkar, Ehsan Rezayat
2026

Source: Crossref; matched terms: visual working memory, working memory, precision, flexibility

Link: <https://doi.org/10.64898/2026.06.28.733231>

Goal: Studying chess experts helps researchers understand how intensive practice shapes thinking skills

Method: Cognitive flexibility is the ability to adjust thoughts when rules or tasks change Working memory is the ability to hold and use information over short periods

Results: Chess players showed better accuracy in both tasks than controls

Conclusion: These results indicate that chess expertise may improve rule-guided flexibility and visual working memory precision in adolescents.

2. Comparative Review of Artificial Intelligence Driven Deep Learning in Machine Vision, Healthcare, and Memory Processing Applications

M. Manikandan, M. Keerthana
2025 International Conference on Data Science, Agents & Artificial Intelligence (ICDSAAI), 2025

Source: Semantic Scholar; matched terms: neural network, neural networks, deep learning, artificial intelligence

Link: <https://www.semanticscholar.org/paper/0a8e3da256aa788e7ed78a30202149f4ac92371d>

Goal: A comparative analysis of three high-impact domains are machine vision and robotic vision, healthcare prediction systems, and memory processing for deep neural networks (DNNs), which have the advancements of deep learning and machine learning

Method: A comparative analysis of three high-impact domains are machine vision and robotic vision, healthcare prediction systems, and memory processing for deep neural networks (DNNs), which have the advancements of deep learning and machine learning

Results: Meanwhile, to address the traditional memory bottlenecks in DNN processing, an Artificial intelligence-driven technology called processing-in-memory (PIM) can be used, resulting in faster computation and increased energy efficiency

Conclusion: The novelty of this work depends on its interdisciplinary approach, bringing together advances in machine vision, healthcare, and PIM to address common challenges like dataflow management, interpretation of models, and energy efficiency It mainly focuses on cross-domain solutions and emerging trends, providing a unique viewpoint on long-term and scalable AI applications across a wide range of fields.

3. Sequence structure organizes items in varied latent states of working memory neural network

Qiaoli Huang, Huihui Zhang, Huan Luo
eLife, 2021

Source: Crossref; matched terms: working memory, computational model, modeling, neural network

Link: <https://doi.org/10.7554/elife.67589>

Goal: In memory experiences, events do not exist independently but are linked with each other via structure-based organization

Method: Here, we combined magnetoencephalogram (MEG) recordings, computational modeling, and impulse-response approaches to probe the latent states when subjects held a list of items in working memory (WM)

Results: Computational modeling further reveals a dominant function of sequence context, instead of passive memory decaying, in characterizing recency effect Taken together, sequence structure context shapes the way WM items are stored in the human brain and essentially influences memory behavior.

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4. Supplemental Material for Adaptive Visual Working Memory: Expecting A Delayed Estimation Task Enhances Visual Working Memory Precision

Authors unavailable

Journal of Experimental Psychology: Human Perception and Performance, 2023

Source: Crossref; matched terms: visual working memory, working memory, precision

Link: <https://doi.org/10.1037/xhp0001066.supp>

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

5. Target familiarity and visual working memory do not influence familiarity effect in visual search

Zhihan Guo, Maolong Niu, Qi Wang

Scientific Reports, 2021

Source: Crossref; matched terms: visual working memory, visual search, working memory

Link: <https://doi.org/10.1038/s41598-021-86669-2>

Goal: Abstract Familiarity effect refers to the phenomenon that searching for a novel target among familiar distractors is more efficient than that searching for a familiar target among novel distractors

Method: Experiment 1 using two similar Chinese characters (“■” and “■”) suggested that searching for a familiar target from familiar distractors is an inefficient search process in Chinese context

Results: Results demonstrated no difference in the search efficiency between single and dual-task conditions

Conclusion: Experiment 1 using two similar Chinese characters (“■” and “■”) suggested that searching for a familiar target from familiar distractors is an inefficient search process in Chinese context